

LAST MOUNTAIN, SK, Canada

WMO: 715560

Lat: 51.415N

Lon: 105.243W

Elev: 1631

StdP: 13.85

Time Zone: -6.00 (W06)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-22.1	-33.9	0.9	-27.5	-27.9	1.2	-22.0	27.8	18.6	25.5	18.2	6.5	190	0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	85.5	66.2	82.0	64.8	78.9	62.9	70.3	79.9	68.0	77.2	65.9	74.9	11.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	106.2	75.1	64.8	97.4	72.3	62.7	90.4	70.3	35.4	80.0	33.4	77.6	31.7	74.8	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.9	19.6	DB	-35.1	92.3	5.5	3.2	-39.1	94.6	-42.3	96.4	-45.4	98.2	-49.4	100.6	(4)
			WB	-35.1	74.8	5.6	2.1	-39.1	76.4	-42.4	77.7	-45.5	78.9	-49.6	80.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	5.9	8.0	20.1	38.3	50.9	59.5	65.1	62.9	53.5	39.1	23.2	9.8	(6)
(7)		DBStd	23.96	15.27	14.46	14.22	10.42	8.40	6.21	5.71	6.42	8.05	8.79	12.46	14.52	(7)
(8)		HDD50	6392	1368	1177	928	365	93	4	0	1	54	352	805	1245	(8)
(9)		HDD65	10557	1833	1597	1393	800	440	185	70	117	353	804	1255	1710	(9)
(10)		CDD50	1464	0	0	0	15	121	289	468	400	158	13	0	0	(10)
(11)		CDD65	155	0	0	0	0	3	20	73	52	7	0	0	0	(11)
(12)		CDH74	2080	0	0	0	6	110	249	831	699	182	3	0	0	(12)
(13)		CDH80	525	0	0	0	2	16	44	225	195	43	0	0	0	(13)
(14)	Wind	WSAvg	9.5	9.6	9.1	9.7	10.5	10.8	9.4	8.8	8.5	9.4	9.7	9.4	9.0	(14)
(15)	Precipitation	PrecAvg	15.9	0.6	0.4	0.5	1.1	1.7	3.2	2.6	2.1	1.5	1.0	0.6	0.6	(15)
(16)		PrecMax	30.0	1.6	0.7	1.3	2.6	5.7	7.0	5.6	9.3	5.6	2.7	2.0	1.6	(16)
(17)		PrecMin	9.4	0.0	0.0	0.0	0.0	0.4	0.6	0.8	0.4	0.0	0.0	0.0	0.1	(17)
(18)		PrecStd	4.8	0.4	0.2	0.4	0.8	1.1	1.6	1.3	1.9	1.1	0.7	0.4	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	39.4	54.7	71.9	82.3	85.3	90.4	90.8	85.1	72.1	54.0	42.3	(19)
(20)			MCWB	35.5	35.5	44.5	51.7	58.4	65.3	69.5	67.4	62.4	54.9	44.8	36.9	(20)
(21)		2%	DB	36.1	35.3	47.5	66.4	77.3	80.0	85.5	85.2	79.5	64.2	46.7	37.4	(21)
(22)			MCWB	33.6	33.2	40.4	49.4	57.0	62.5	68.4	65.5	60.5	51.0	40.5	34.3	(22)
(23)		5%	DB	32.8	32.5	42.0	60.9	72.7	76.5	82.0	81.3	74.0	58.0	42.2	33.6	(23)
(24)			MCWB	31.0	31.0	37.6	47.0	54.6	61.0	66.7	64.5	58.4	47.8	37.9	31.5	(24)
(25)		10%	DB	28.0	28.8	37.4	55.7	68.4	73.4	78.6	77.7	69.1	53.3	38.4	29.4	(25)
(26)			MCWB	26.9	27.5	34.9	44.8	52.6	59.2	65.4	62.9	55.9	45.9	35.7	27.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	36.4	45.3	54.6	62.2	70.3	75.2	71.4	65.8	57.0	45.7	37.4	(27)
(28)			MCDWB	38.5	38.6	53.5	67.8	73.6	79.0	83.2	82.6	78.6	68.1	52.4	41.7	(28)
(29)		2%	WB	33.8	33.4	41.0	51.0	59.2	66.5	71.7	69.0	62.8	52.5	41.1	34.6	(29)
(30)			MCDWB	35.9	35.1	46.6	62.6	71.5	74.9	80.8	79.7	74.5	61.7	45.5	37.3	(30)
(31)		5%	WB	31.0	31.1	37.7	48.5	56.8	63.7	69.3	66.8	60.2	49.4	38.4	31.5	(31)
(32)			MCDWB	32.8	32.5	41.6	58.9	69.7	72.1	78.2	76.8	71.0	56.5	41.6	33.6	(32)
(33)		10%	WB	26.8	27.5	35.1	45.8	54.5	61.6	67.0	64.8	57.5	46.5	35.7	27.8	(33)
(34)			MCDWB	27.9	28.8	37.5	54.5	65.6	69.6	75.5	74.2	67.5	52.9	38.3	29.4	(34)
(35)	Mean Daily Temperature Range	MDBR	19.5	19.6	19.4	22.6	24.7	21.3	22.9	24.5	24.1	20.9	17.1	17.5	(35)	
(36)		5% DB	MCDBR	21.1	19.2	22.0	31.9	32.2	27.3	28.2	30.5	33.1	29.7	21.3	20.1	(36)
(37)			MCWBR	19.2	17.0	15.4	16.5	14.6	13.0	13.5	13.5	16.0	17.2	15.1	17.2	(37)
(38)		5% WB	MCDWB	21.0	18.3	21.3	29.5	27.6	22.7	25.0	25.9	29.4	27.8	20.2	19.9	(38)
(39)			MCWBR	19.1	16.4	15.4	16.2	13.4	12.0	13.6	13.0	14.8	16.7	14.7	17.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.251	0.270	0.286	0.328	0.343	0.353	0.360	0.366	0.318	0.285	0.252	0.237	(40)	
(41)		taud	2.311	2.269	2.398	2.396	2.400	2.416	2.398	2.376	2.492	2.546	2.488	2.365	(41)	
(42)		Ebn at Noon	249	274	292	290	289	286	281	274	278	269	252	239	(42)	
(43)		Edh at Noon	20	27	30	34	35	35	35	34	27	21	16	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	384	719	1182	1528	1870	1912	2010	1620	1198	699	355	271	(44)	
(45)		RadStd	38	50	104	128	158	132	116	113	96	95	50	27	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page